AICTE | IBM SKILLSBUILD · MACHINE LEARNING & APPLIED AI INTERNSHIP 2026

FasalMitra

Smart crop and fertilizer advisor for small farmers

SDG 2 · Zero Hunger

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01 · THE PROBLEM

Small farmers choose crops and fertilizer by guesswork

Guesswork

Crop and fertilizer choices often follow habit or a neighbour's advice, not what the soil and weather support.

Wasted money

Too much fertilizer costs money and does not raise yield. The wrong crop for the land lowers income.

Unread soil data

Soil Health Cards list N, P, K and pH values, but few farmers can turn them into a decision.

Linked goals: **SDG 2** Zero Hunger · **SDG 12** Responsible Consumption · **SDG 13** Climate Action

02 · WHO IT HELPS

Farmers first, and the people who support them

Primary users

- Small and marginal farmers
- Farmers who hold a Soil Health Card but have no advisor nearby

Early adopters

- Farmer groups and producer organisations
- Krishi Vigyan Kendras and extension workers
- NGOs and college agri clubs working with villages

Available in **English and Hindi**, so language is not a barrier.

03 · THE SOLUTION

From a soil-test card to a clear plan in three steps

1

Enter field data

Seven numbers: N, P, K, pH, temperature, humidity and rainfall. One tap fills an example.

2

The model ranks crops

A trained classifier scores 22 crops and shows the top three with a suitability percentage.

3

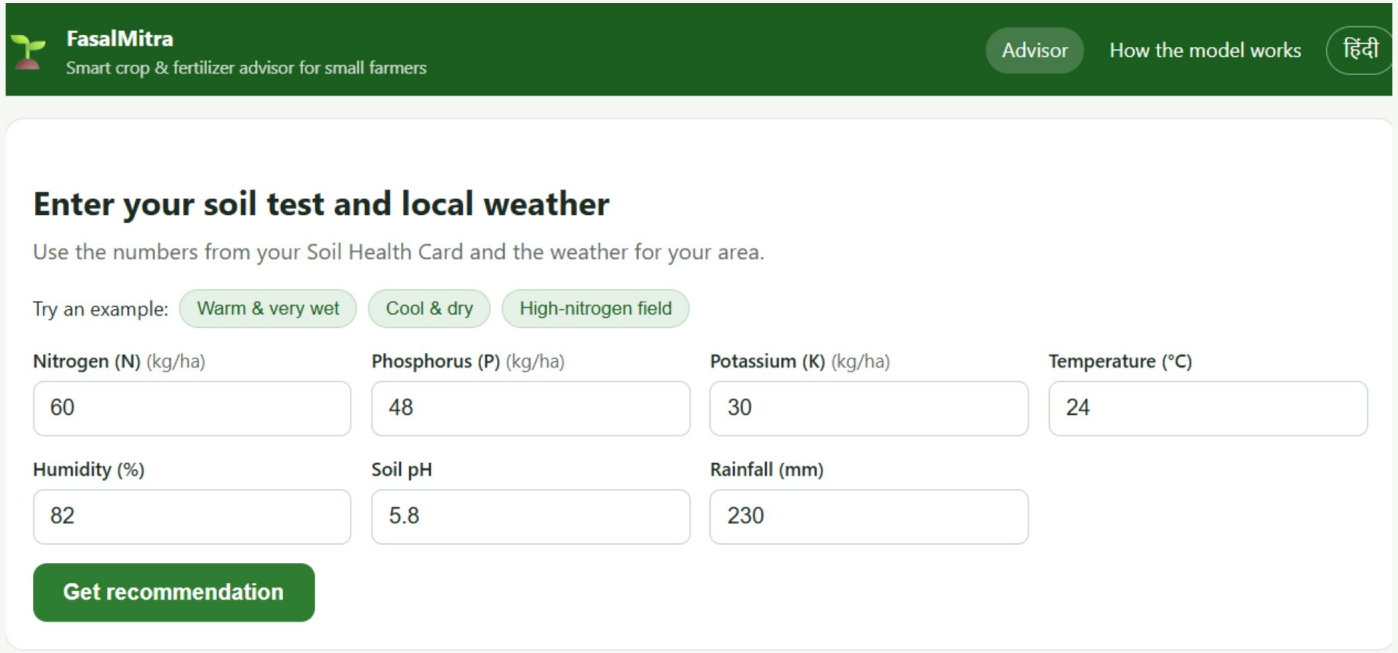
Get a plain plan

For each crop: which values match, what fertilizer to add, and what to skip to save money.

Runs in any phone browser · built with Python, scikit-learn and Flask

04 · LIVE DEMO

See it working



- Type the soil-test values, or tap a ready example
- Get the top three crops with a suitability score
- Switch to Hindi with one tap

Every recommendation shows its reasons

WHY THIS CROP?

▼	Nitrogen (N)	**60.0**	67.41–90.32
✓	Phosphorus (P)	**48.0**	39.41–58.52
▼	Potassium (K)	**30.0**	32.26–48.06
✓	Temperature (°C)	**24.0**	21.47–26.69
✓	Humidity (%)	**82.0**	74.62–87.53
▼	Soil pH	**5.8**	5.89–6.87
✓	Rainfall (mm)	**230.0**	172.01–289.27

FERTILIZER PLAN

N is low by about 20 kg/ha. Add about 45 kg/ha of urea.

P is adequate. No extra needed.

K is low by about 10 kg/ha. Add about 15 kg/ha of muriate of potash (MOP).

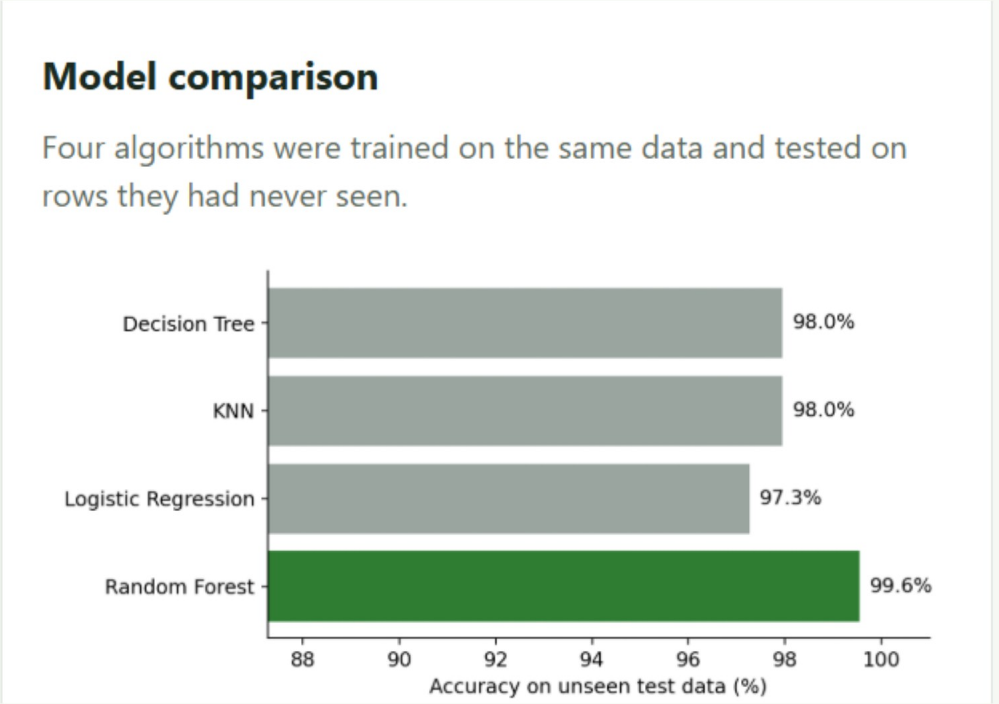
Soil pH suits this crop.

How the advice works

- Compare each nutrient with the crop's typical level in the data
- Well below it: add. Well above it: skip and save money
- Quantities are estimates; confirm with a KVK

Four models, one winner

1. **Get data:** soil and weather values labelled with the crop
2. **Prepare:** clean column names, remove duplicates
3. **Train four models:** Decision Tree, KNN, Logistic Regression, Random Forest
4. **Test:** 20% of rows held back, plus 5-fold cross-validation
5. **Keep the best** and use it in the web app



07 · IMPACT

How we would measure success

Farmers advised

Completed recommendations by real users

Fertilizer saved

Kilograms per hectare not applied after a “skip” advice

Advice accepted

Share of plans a local agri officer agrees with

Repeat use

Farmers who come back the next season

SDG 2: better-matched crops, steadier yields

SDG 12: less fertilizer wasted

SDG 13: less over-application, so less run-off and fewer emissions

Free for farmers, funded by partners

Revenue

- Licences for NGOs, producer groups and CSR programmes
- Partnerships with agri departments and KVKs
- No fertilizer-brand sponsorship, so advice stays unbiased

Channels

- KVKs and extension workers
- WhatsApp sharing and farmer groups
- College agri clubs and student ambassadors

Costs

- Hosting and maintenance
- Translation and field validation with agronomists
- Data updates and user support

What it cannot do yet, and what comes next

Honest limits

- Trained on a general benchmark dataset, not local farm records
- Fertilizer quantities are estimates, not agronomist-validated
- No yield or price prediction yet

Next steps

- Add local soil data and validate with agronomists
- Predict yield with regression
- WhatsApp and voice access in more languages

Thank you

Questions and feedback are welcome

Demo: runs locally · Code: https://github.com/ShreyashSuryaDubey/FasalMitra

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